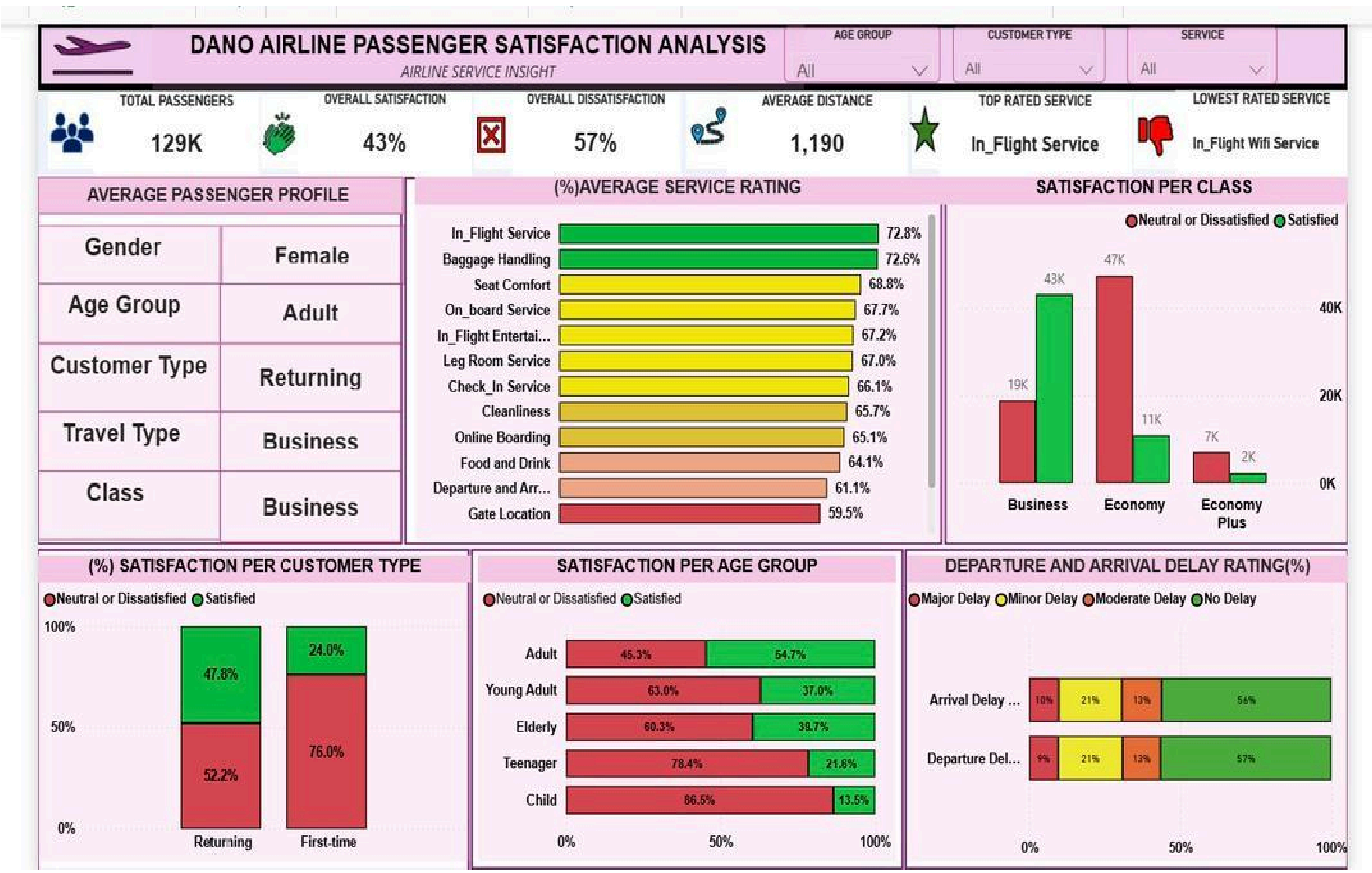
DANO AIRLINE DATA DASHBOARD



Airline Passenger Satisfaction Analysis Report

Introduction

This report presents a comprehensive analysis of the Dano Airline Passenger Satisfaction dataset. The primary objective of the analysis is to investigate the factors contributing to customer satisfaction and dissatisfaction, identify patterns in passenger experiences, and provide actionable insights that can support airline management in improving service quality and overall customer satisfaction.

The analysis was conducted using Microsoft Power BI. The dataset underwent extensive data cleaning, transformation, modelling, and visualization to produce an interactive dashboard capable of answering key business questions and highlighting the major drivers of passenger satisfaction.

Data Overview

The dataset contains detailed information on airline passengers, their travel characteristics, flight performance, and their ratings of various airline services.

The dataset includes information such as:

- Passenger ID
- Gender
- Customer Type
- Age
- Type of Travel
- Class
- Flight Distance
- Departure Delay
- Arrival Delay
- Departure and Arrival Time Convenience
- Ease of Online Booking
- Gate Location
- Food and Drink
- Online Boarding
- Seat Comfort
- Inflight Entertainment
- On Board Service
- Leg Room Service
- Baggage Handling
- Check In Service
- Inflight Service

- Cleanliness
- Satisfaction Level

The dataset provides a comprehensive view of passenger demographics, travel behaviour, operational performance, and service quality, making it suitable for analysing the key drivers of customer satisfaction.

Data Cleaning and Preparation

To ensure the accuracy, consistency, and reliability of the analysis, the dataset was cleaned and transformed using Power BI Power Query. The following data preparation activities were performed.

Handling Missing Values

The Arrival Delay column contained missing values. Since arrival delays are operational outcomes that cannot be accurately inferred from departure delays, records containing missing arrival delay values were removed to preserve data integrity and avoid introducing bias into the analysis.

Age Standardization

The numerical **Age** column was grouped into five standardized categories to facilitate demographic analysis.

- Child (0–12 years)
- Teenager (13–19 years)
- Young Adult (20–35 years)
- Adult (36–55 years)
- Elderly (56 years and above)

Departure Delay Standardization

The Departure and Arrival Delay columns were transformed into standardized delay categories based on operationally meaningful thresholds.

- No Delay (0 minutes)
- Minor Delay (1–15 minutes)
- Moderate Delay (16–45 minutes)
- Major Delay (More than 45 minutes)

These categories simplify the interpretation of operational performance and enable comparative analysis of delay severity.

Service Rating Standardization

Passenger service ratings recorded on a scale of 0–5 were standardized into meaningful descriptive categories to improve readability and interpretation.

For services containing values **0–5**:

- 0 = Not Applicable
- 1 = Very Poor
- 2 = Poor
- 3 = Average
- 4 = Good
- 5 = Excellent

This categorization was applied to variables including:

- Departure and Arrival Time Convenience
- Ease of Online Booking

For service variables recorded on a **1–5** scale (without a value of 0), the following categories were used:

- 1 = Very Poor
- 2 = Poor
- 3 = Average
- 4 = Good
- 5 = Excellent

This categorization was applied to:

- Baggage Handling
- Seat Comfort
- Food and Drink
- Online Boarding
- Inflight Entertainment
- On Board Service
- Leg Room Service
- Inflight Service
- Check In Service
- Cleanliness
- Gate Location

Data Type Validation

All columns were assigned appropriate data types to ensure accurate calculations, aggregations, filtering, and visualization within Power BI.

These cleaning and transformation steps improved data consistency, enhanced analytical reliability, and established a standardized dataset suitable for dashboard development and business intelligence analysis.

Data Modelling

Following data cleaning and transformation, a star schema data model was developed in Power BI to improve analytical efficiency, reduce data redundancy, and support interactive dashboard reporting. The model separates passenger demographic information from travel related attributes while maintaining a centralized fact table containing passenger level records.

The data model consists of **one fact table** and **two dimension tables**.

Fact Table

Fact_Airline

The **Fact Airline** table serves as the central table of the data model. Each record represents an individual passenger and contains operational, service quality, and satisfaction information used for analysis.

The table includes attributes such as:

- Passenger ID
- Flight Distance
- Departure Delay
- Arrival Delay
- Departure Delay Category
- Arrival Delay Category
- Satisfaction
- Service ratings

This table provides the quantitative data used in calculations, KPIs, and dashboard visualizations.

Dimension Tables

Dim_Customer

The **Dim Customer** table contains passenger demographic information.

Attributes include:

- Passenger ID
- Gender
- Age
- Age Group
- Customer Type

This table supports demographic analysis of passenger satisfaction and customer segmentation.

Dim Travel

The Dim Travel table contains passenger travel characteristics.

Attributes include:

- Passenger ID
- Type of Travel
- Class
- All services

This table enables analysis of passenger satisfaction across different travel purposes and cabin classes.

Relationships

Relationships were established between the Fact_Airline table and both dimension tables using the Passenger ID field.

This star schema allows filters and slicers to propagate efficiently across the model while maintaining data integrity and improving query performance.

The resulting model provides a structured framework for analysing passenger demographics, travel behaviour, operational performance, service quality, and customer satisfaction.

Measures and Calculations

The following DAX measures and calculations were created to support the dashboard analysis.

Total Passengers

Calculates the total number of passengers contained in the dataset. This measure provides the basis for all percentage calculations.

Overall Satisfaction Rate

Calculates the percentage of passengers classified as **Satisfied**.

This KPI measures the proportion of passengers who reported a satisfactory travel experience.

Overall Dissatisfaction Rate

Calculates the percentage of passengers classified as **Neutral or Dissatisfied**.

This measure complements the Overall Satisfaction Rate and highlights the percentage of passengers whose expectations were not fully met.

Average Flight Distance

Calculates the average flight distance travelled by passengers.

This metric provides insight into the typical journey length represented in the dataset.

Top Rated Service

Identifies the airline service with the highest average passenger rating across all evaluated service attributes. This KPI highlights the airline's strongest performing service area.

Lowest Rated Service

Identifies the airline service receiving the lowest average passenger rating.

This measure helps identify the service requiring the greatest improvement.

Typical Passenger Profile

Measures were created to determine the most frequently occurring passenger characteristics within the dataset, including:

- Gender
- Age Group
- Customer Type
- Type of Travel
- Class

These measures dynamically update when dashboard filters are applied, allowing the dashboard to display the typical passenger profile for the selected data.

Average Service Rating (%)

Calculates the average rating for each airline service and converts the result into a percentage.

This measure was used to compare service performance across all service attributes, allowing the identification of highly rated and poorly rated services.

Satisfaction by Class

Calculates the distribution of satisfied and neutral or dissatisfied passengers across Business, Economy, and Economy Plus classes.

This measure supports comparison of customer satisfaction across different cabin classes.

Satisfaction by Customer Type

Calculates the percentage distribution of satisfied and neutral or dissatisfied passengers among Returning and First time customers.

This measure helps determine whether customer loyalty influences satisfaction.

Satisfaction by Age Group

Calculates the percentage distribution of satisfied and neutral or dissatisfied passengers across the different age groups.

This allows management to identify demographic groups experiencing lower levels of satisfaction.

Departure and Arrival Delay Rating Distribution (%)

Calculates the percentage distribution of flights across the four delay categories:

- No Delay
- Minor Delay
- Moderate Delay
- Major Delay

The measure compares departure and arrival operational performance and highlights the proportion of flights experiencing each delay category.

Dashboard Overview

An interactive dashboard was developed in Microsoft Power BI to provide a comprehensive analysis of airline passenger satisfaction. The dashboard consolidates key performance indicators (KPIs), customer demographic information, service quality metrics, and operational performance into a single interface, enabling users to monitor performance and investigate the factors influencing passenger satisfaction.

The dashboard incorporates interactive slicers for Age Group, Customer Type, and Service, allowing users to dynamically filter the analysis and explore satisfaction trends across different passenger segments. The dashboard is designed to support evidence based decision making by presenting both high level performance metrics and detailed analytical visualizations.

The dashboard consists of KPI cards, comparative bar charts, stacked bar charts, and summary tables that collectively provide insights into passenger satisfaction, customer demographics, service performance, and flight delays.

Dashboard Objectives

The dashboard was designed to achieve the following objectives:

- Evaluate the overall level of passenger satisfaction.
- Compare satisfaction and dissatisfaction rates across different passenger groups.
- Identify the airline services that receive the highest and lowest passenger ratings.
- Examine customer satisfaction across different travel classes.
- Analyse satisfaction patterns among customer types and age groups.
- Evaluate operational performance through departure and arrival delay distributions.
- Identify the characteristics of the airline's typical passenger.
- Provide management with actionable insights to improve customer experience and operational efficiency.

Key Performance Indicators (KPIs)

The dashboard contains the following Key Performance Indicators to support business analysis:

KPI	Description
Total Passengers	Displays the total number of passengers included in the analysis.
Overall Satisfaction Rate	Shows the percentage of passengers who reported being satisfied.
Overall Dissatisfaction Rate	Displays the percentage of passengers who were neutral or dissatisfied.
Average Flight Distance	Indicates the average flight distance travelled by passengers.
Top Rated Service	Identifies the service with the highest average passenger rating.
Lowest Rated Service	Identifies the service with the lowest average passenger rating.
Typical Passenger Profile	Displays the most common passenger characteristics, including gender, age group, customer type, travel type, and class.
Average Service Rating (%)	Compares the average percentage rating across all airline services.

Satisfaction by Class	Compares satisfaction levels across Business, Economy, and Economy Plus passengers.
Satisfaction by Customer Type	Compares satisfaction between Returning and First time customers.
Satisfaction by Age Group	Examines satisfaction levels across different age groups.
Departure and Arrival Delay Rating (%)	Compares the distribution of departure and arrival delays across standardized delay categories.

Dashboard Visualizations

The dashboard comprises several visualizations designed to present information clearly and support effective decision making.

KPI Cards

Six KPI cards are positioned at the top of the dashboard to provide a high-level summary of airline performance. These display the total number of passengers, overall satisfaction rate, overall dissatisfaction rate, average flight distance, top rated service, and lowest rated service.

Typical Passenger Profile Table

A summary table displays the most frequently occurring passenger characteristics, including gender, age group, customer type, travel type, and travel class. This visualization provides a quick overview of the airline's predominant customer segment.

Average Service Rating (%)

A horizontal bar chart compares the average percentage rating of each airline service. This visualization highlights services that consistently perform well while identifying services that require improvement.

Satisfaction by Class

A clustered column chart compares the number of satisfied and neutral or dissatisfied passengers across Business, Economy, and Economy Plus travel classes. This visualization illustrates how satisfaction varies across cabin classes.

Satisfaction by Customer Type

A 100% stacked column chart compares the percentage of satisfied and neutral or dissatisfied passengers between Returning and First time customers, enabling assessment of customer loyalty and satisfaction.

Satisfaction by Age Group

A 100% stacked bar chart displays the percentage distribution of satisfaction across different age groups. This visualization helps identify demographic groups with relatively lower satisfaction levels.

Departure and Arrival Delay Rating (%)

A 100% stacked bar chart compares the percentage distribution of departure and arrival delays across four delay categories: No Delay, Minor Delay, Moderate Delay, and Major Delay. This visualization provides insight into operational efficiency and delay patterns.

Dashboard Findings and Analysis

The dashboard provides valuable insights into passenger satisfaction, service quality, customer demographics, and operational performance. The findings are discussed below.

Overall Passenger Satisfaction

The analysis revealed that only **43%** of passengers reported being satisfied with their travel experience, while **57%** were classified as neutral or dissatisfied.

This indicates that the airline is experiencing a relatively low level of customer satisfaction, with more than half of its passengers expressing dissatisfaction or failing to report a satisfactory experience. This finding suggests that improvements in service quality and operational efficiency are required to enhance the overall customer experience.

Passenger Profile

The dashboard identified the airline's typical passenger profile as follows:

- Gender: Female
- Age Group: Adult
- Customer Type: Returning
- Travel Type: Business
- Travel Class: Business

These findings suggest that the airline primarily serves adult business travellers who are repeat customers and predominantly travel in Business Class. Since returning customers represent the largest customer segment, maintaining their satisfaction is essential for customer retention and long term business growth.

Service Quality Analysis

The Average Service Rating chart indicates considerable variation in passenger perceptions of airline services.

The highest rated services include:

- Inflight Service (72.8%)
- Baggage Handling (72.6%)
- Seat Comfort (68.8%)

These services appear to meet passenger expectations and represent operational strengths that should be maintained.

Conversely, the lowest rated services include:

- Gate Location (59.5%)
- Departure and Arrival Time Convenience (61.1%)
- Food and Drink (64.1%)

Although these services were not rated extremely poorly, their comparatively lower ratings suggest opportunities for improvement. In particular, Inflight WiFi Service was identified as the overall lowest rated service, indicating that passengers may experience connectivity challenges or perceive poor value from the onboard internet service.

Overall, service ratings demonstrate that while onboard services generally perform well, improvements in convenience related services could significantly enhance passenger satisfaction.

Satisfaction by Travel Class

Passenger satisfaction varied considerably across travel classes.

Business Class recorded the highest number of satisfied passengers, with approximately **43,000** satisfied passengers compared to **19,000** neutral or dissatisfied passengers.

Economy Class displayed the opposite trend, with approximately **47,000** passengers classified as neutral or dissatisfied compared to only **11,000** satisfied passengers.

Economy Plus also recorded more dissatisfied passengers than satisfied passengers, although the passenger volume was considerably smaller.

These findings suggest that passengers travelling in premium cabins generally experience higher service satisfaction than those travelling in Economy Class. This highlights Economy Class as the primary area requiring service improvements.

Satisfaction by Customer Type

The dashboard revealed notable differences between customer types.

Among Returning customers:

- 47.8% were satisfied.
- 52.2% were neutral or dissatisfied.

Among First time customers:

- Only 24.0% were satisfied.
- 76.0% were neutral or dissatisfied.

This indicates that first time passengers are substantially less satisfied than returning customers. Since first impressions strongly influence customer loyalty, improving the experience of first time travellers may increase customer retention and future repeat business.

Satisfaction by Age Group

Passenger satisfaction varied significantly across age groups.

Adults recorded the highest satisfaction level, with approximately **54.7%** satisfied.

Young Adults showed a relatively low satisfaction rate of **37.0%**, while Elderly passengers recorded **39.7%** satisfaction.

Teenagers and Children exhibited the lowest satisfaction levels, with satisfaction rates of **21.6%** and **13.5%**, respectively.

These findings suggest that younger passengers experience lower levels of satisfaction than adults. Although children may not directly evaluate airline services themselves, their responses likely reflect the travel experiences of accompanying families.

Operational Performance

The analysis of departure and arrival delay distributions indicates that operational performance is relatively consistent.

Approximately **57%** of departures experienced no delay, while approximately **56%** of arrivals arrived without delay.

Minor delays accounted for roughly **21%** of both departures and arrivals.

Moderate delays represented approximately **13%**, while major delays accounted for only **9–10%** of flights.

Although the airline demonstrates relatively good operational performance, the presence of moderate and major delays affecting nearly one quarter of flights may still negatively influence passenger satisfaction, particularly among business travellers who place greater importance on punctuality.

Recommendations

Based on the findings, the following recommendations are proposed to improve passenger satisfaction and overall service quality.

Improve Inflight WiFi Services

Inflight WiFi received the lowest overall service rating. The airline should evaluate internet speed, reliability, pricing, and coverage to improve the onboard connectivity experience.

Enhance Economy Class Experience

Economy Class recorded substantially lower satisfaction levels than Business Class. Improving seat comfort, food quality, boarding efficiency, and customer service within Economy Class may significantly improve overall passenger satisfaction.

Focus on First Time Passengers

First time travellers exhibited considerably lower satisfaction levels than returning customers. The airline should implement initiatives such as improved onboarding, clearer travel information, and personalised customer support to create stronger first impressions.

Improve Convenience Related Services

Lower ratings for gate location and departure and arrival time convenience suggest opportunities to improve airport navigation, passenger communication, and scheduling efficiency.

Reduce Operational Delays

Although most flights departed and arrived on time, reducing moderate and major delays could further improve customer satisfaction, particularly for business travellers whose schedules are more time sensitive.

Maintain High Performing Services

Services such as Inflight Service, Baggage Handling, and Seat Comfort received relatively high ratings and should continue to receive adequate investment to sustain customer satisfaction.

Responses to the Capstone Business Questions

1. What is the overall passenger satisfaction level?

Overall passenger satisfaction stands at **43%**, while **57%** of passengers were classified as neutral or dissatisfied. This indicates that the airline faces significant challenges in meeting passenger expectations.

2. Which customer segments exhibit the highest and lowest satisfaction?

Business Class passengers and Adults demonstrate the highest satisfaction levels.

First time customers, Children, and Teenagers exhibit the lowest satisfaction levels.

3. Which airline services require improvement?

Inflight WiFi Service, Gate Location, Departure and Arrival Time Convenience, and Food and Drink received comparatively lower ratings and should be prioritised for service improvement initiatives.

4. How does operational performance affect passenger experience?

Although over half of flights experienced no delays, approximately 43–44% experienced some form of delay. Reducing operational delays may contribute to higher passenger satisfaction, particularly among business travellers.

5. What actions should management prioritise?

Management should prioritise:

- Improving Economy Class services.
- Enhancing Inflight WiFi quality.
- Improving first time passenger experiences.
- Reducing operational delays.
- Maintaining high performing services while addressing weaker service areas.

Conclusion

This analysis provided a comprehensive assessment of airline passenger satisfaction by integrating passenger demographics, travel characteristics, service quality ratings, and operational performance into an interactive Power BI dashboard.

The findings revealed that overall satisfaction remains relatively low, with only **43%** of passengers reporting satisfaction. Service quality varies across operational areas, with Inflight Service and Baggage Handling performing strongly, while Inflight WiFi and convenience related services require improvement. Satisfaction also differs substantially across travel classes, customer types, and age groups, with Economy Class passengers and first time travellers exhibiting particularly low satisfaction levels.

The dashboard provides management with an effective decision support tool for monitoring passenger experiences, identifying performance gaps, and prioritising service improvements. By implementing the recommendations outlined in this report, the airline can enhance customer satisfaction, strengthen passenger loyalty, and improve its overall competitive position.